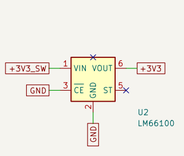
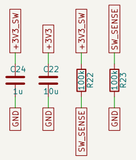
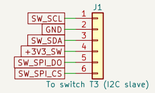
POWER
The board runs from the switch's 3.3 V (T3/T8 pin 4) through the ideal diode U2. On the bench a debug probe can supply 3.3 V on J5 pin 1 instead; U2 keeps it out of the switch. SW_SENSE tells the firmware whether the switch is on: until it is, keep every switch-facing pin high-impedance with its input buffer off.

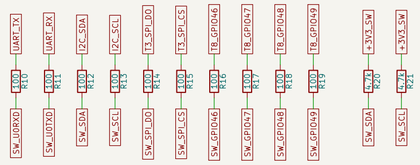




MCU
Default build: RP2350A with U5, a W25Q128JVSIQ (16 MiB).
Alternative: RP2354A (2 MiB in the package, same pinout) with U5 and C19 left off.
Never fit U5 with an RP2354A: both flash chips would answer on QSPI_SS.

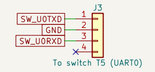
SWITCH HEADERS (J1–J3 are male headers on the board's underside)
100R in series on every switch-facing line: I2C pull-ups to the switch's 3.3 V.
J3 pin 4 (the switch's 3.3 V on T5) is left open.

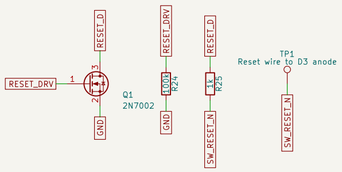






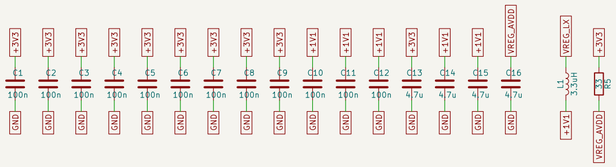
SOC RESET
TP1 goes to D3's ANODE on the switch: the front pad, by the "+".
The banded end is the 3.3 V rail. R25 limits the current if the wire is misplaced.

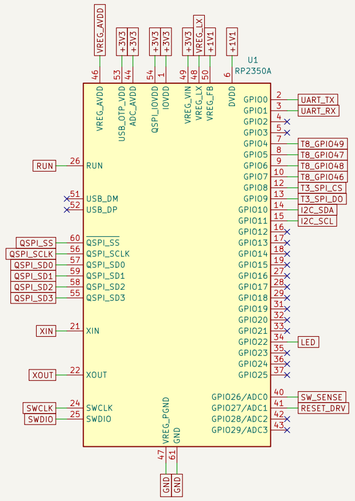




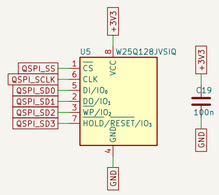


CORE REGULATOR AND DECOUPLING
L1, C13 (CIN), C14 (COUT), R5/C16 (VREG_AVDD filter): lay out as the Pico 2 does (datasheet 6.3.8).
C15 is the second 4.7u at DVDD pin 23. One 100n per IOVDD and DVDD pin, at the pin.

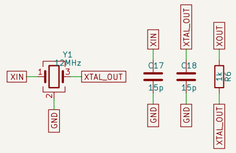


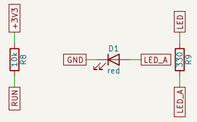


FLASH, CLOCK, DEBUG
U5/C19: fitted with the RP2350A, left off with an RP2354A.
Programming and debug are over SWD (J5); there is no USB.









Generated by hardware/make_sch.py: not built or tested yet
For SWTG024AS-V2.0 switches: management processor on headers T3, T8 and T5
github.com/wrcrooks/rtlplayground-modchip

Sheet: /
File: modchip.kicad_sch

Title: RTLPlayground mod chip (RP2350)

Size: A2	Date:	**Rev: A0**
KiCad E.D.A. 10.0.5		Id: 1/1